

MIGRATION PROGRAM PLAN

Migration of 20 .NET / Java Applications to Google Cloud Platform (GCP)

1. Executive Summary

Objective

Migrate 20 .NET / Java applications and their associated PostgreSQL databases from two legacy platforms to Google Cloud Platform (GCP) within 6 months. The migration will be executed using a Migration Factory Model, where a secure cloud foundation is built once and application migrations are executed in repeatable waves.

Business Goals

The program aims to:

- Improve platform resiliency
- Achieve high availability (HA)
- Implement disaster recovery (DR)
- Improve latency and throughput
- Reduce infrastructure management overhead
- Retire legacy platform operating costs
- Improve deployment speed through automation

Success Criteria

By Month 6:

- All 20 applications running in GCP
 - All databases migrated successfully
 - Zero data loss during cutovers
 - Defined SLOs achieved
 - HA and DR validated through testing
 - Legacy infrastructure fully decommissioned
 - Operational ownership transitioned to SRE
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2. Key Assumptions

Area	Assumption
Current State	Applications hosted on legacy on-prem/private datacenter platforms
Applications	10 .NET and 10 Java applications
Databases	PostgreSQL databases totaling tens of TB
CI/CD	Existing pipelines available and reusable
Security	Financial-services-grade compliance controls required
Team	Dedicated Platform, App, Security, DBA, QA and SRE teams available
Budget	Funding approved for migration program
Timeline	6-month migration deadline
Cloud	GCP tenant available at project kickoff

3. Application Portfolio Assessment

Applications will be categorized according to business criticality.

Tier	Description	Count
Tier 1	Revenue-critical and customer-facing applications	5
Tier 2	Business-critical applications	8
Tier 3	Internal and lower-risk applications	7

Migration Sequencing Principles

1. Business Criticality
2. Technical and Data dependencies
3. Migration complexity
4. Operational readiness

The Pilot wave will primarily consist of Tier-3 applications and the subsequent migration waves will follow dependency analysis rather than tiering alone.

4. Target State Overview

Cloud Foundation

A secure GCP landing zone will provide:

- Networking
- Identity and access management
- Security controls
- Observability and centralized monitoring
- CI/CD pipelines
- Cost management

Application Platform

Applications will be:

- Containerized where feasible
- Deployed to a highly available cloud platform
- Automated through CI/CD pipelines
- Monitored using centralized observability tools

Database Platform

PostgreSQL databases will be migrated to managed cloud database services with:

- High Availability
- Automated Backups
- Point-In-Time Recovery
- Cross-region Disaster Recovery

Reliability Targets

KPI	Target
Availability	99.95%
RPO	<= 5 Minutes
RTO	<= 60 Minutes
Scalability	Autoscaling Enabled
DR Validation	Quarterly DR Testing

5. Migration Strategy

Application Disposition

Strategy	Description	Expected Apps
Replatform	Containerize and migrate with minimal changes	15
Refactor	Limited modernization for complex applications	3
Rehost	Lift-and-shift where modernization isn't feasible	2

Migration Workflow

Discovery → Dependency Mapping → 6Rs Assessment → Containerization → Testing → Database Migration → Cutover Rehearsal → Production Cutover → Hypercare → SRE Handoff

Migration Waves

Pilot Wave (3 applications)

- 1 .NET application
- 1 Java application
- 1 Batch application

Objective:

- Validate tooling
- Validate deployment process
- Validate rollback procedures

Wave 1 (5 applications)

Focus:

- Low complexity
- Independent services

Wave 2 (6 applications)

Focus:

- Complex applications
- Shared databases
- Legacy workloads

Wave 3 (6 applications)

Focus:

- Remaining applications
- Final production cutovers

6. Dependency Management Framework

One of the largest risks in cloud migrations is unmanaged dependencies.

Technical Dependencies

- Shared databases
- APIs
- Middleware
- Authentication systems
- File transfers

Organizational Dependencies

- Security approvals
- Environment provisioning
- Change management approvals

External Dependencies

- Third-party vendors
- Regulatory dependencies
- Network connectivity

Governance

The TPM will conduct:

- Weekly dependency review board
- Dependency dashboard
- Escalation management
- Critical-path tracking

7. Program Operating Model

Governance Structure

Executive Steering Committee (Frequency: Weekly)

Participants:

- Executive Sponsor
- Platform Leads
- Security Lead
- Application Owners
- SRE Lead
- TPM

Responsibilities:

- Remove blockers and Escalations
- Approve major decisions
- Approve go/no-go milestones

Workstreams

Workstream	Lead
Program Management	TPM
Platform Foundation	Platform Lead
Application Migration	Application Teams
Database Migration	DBA Lead
Testing	QA Lead
Security	Security Lead
Business Readiness	Change Manager
Cutover & Hypercare	SRE Lead

RAID Governance

The TPM will maintain a centralized RAID log throughout the lifecycle of the program.

The RAID Log tracks:

- Risks
- Assumptions
- Issues
- Dependencies

Process

- Weekly review meetings
- Executive escalation of Red items
- Steering committee visibility
- Continuous monitoring

The RAID Log is considered the system of record for program execution.

8. Business Readiness Workstream

Technical success alone does not guarantee program success.

Communication

- Business stakeholder updates
- User notifications
- Executive reporting

Training

- Support teams
- Operations teams
- Application owners

Documentation

- Runbooks
- Standard Operating Procedures
- Escalation procedures

Operational Readiness

- Support model validation
 - Incident-management process validation
 - DR preparedness validation
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9. Change Management Strategy

All production migrations will follow formal change-management controls.

Pre-Cutover

- CAB approval
- Change request approval
- Production readiness review
- Rollback validation

During Cutover

- Dedicated war room
- Executive communication bridge
- Real-time monitoring
- Decision log

Post-Cutover

- Hypercare support
 - Incident review
 - Change closure
 - Lessons learned
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10. Resource Model

	Role	Staffing
TPM		1
Platform Engineers		2-3
Cloud Architects		1-2
DBA Engineers		1-2
Security Engineers		1
QA Engineers		2
SRE Engineers		2
Application SMEs		Per Application

Resource Principles

- Dedicated migration capacity
- Shared ownership across teams
- Early SRE engagement
- Backfill for critical SMEs

11. Communication Strategy

Audience	Frequency	Purpose
Workstreams	Daily	Execution Tracking
Program Team	Weekly	Status Review
Steering Committee	Weekly	Governance
Stakeholders	Monthly	Progress Updates
Executives	Monthly	Strategic Reporting

Communication Artifacts

- Weekly Status Report
- RAID Dashboard
- Executive Dashboard
- Cutover Communications
- Steering Committee Package

12. Six-Month Roadmap

Month	Milestone	Deliverable
Month 1	Foundation Ready	Landing zone and migration planning complete
Month 2	Pilot Proven	3 pilot applications migrated
Month 3	Wave 1 Complete	5 applications migrated
Month 4	Wave 2 Complete	6 applications migrated
Month 5	Wave 3 Complete	Remaining applications migrated
Month 6	Migration Complete	Hypercare, DR validation and decommissioning complete

13. Key Decisions and Approval Gates

Gate	Approver
Foundation Ready	Security + Platform
Pilot Go-Live	Steering Committee
Production Cutover	CAB
DR Validation	Infrastructure Leadership
Legacy Decommission	Executive Sponsor

Exit Criteria

Each gate requires:

- Test completion
- Rollback validation
- Change approval
- Stakeholder sign-off

14. Milestone Gates

M1: Foundation Ready

Exit Criteria:

- Landing zone approved
- Security sign-off obtained
- Application inventory completed
- Dependency map completed
- Wave plan approved

M2: Pilot Proven

Exit Criteria:

- Pilot applications successfully migrated
- SLOs validated
- Migration runbooks completed
- Rollback tested
- DR test passed

M3-M5: Wave Completion

Exit Criteria:

- Applications migrated
- Performance validated
- No unresolved Sev-1 or Sev-2 issues
- Hypercare completed

M6: Migration Complete

Exit Criteria:

- All applications in GCP
- DR validated
- SLOs green
- Legacy infrastructure decommissioned
- SRE handoff completed

15. Testing Strategy

Testing occurs for every migration wave.

Functional Testing

- Application functionality validation
- Regression testing

Data Validation

- Record counts
- Checksums
- Data reconciliation

Performance Testing

- Load testing
- Stress testing
- Latency validation

Security Testing

- Vulnerability scans
- Secret validation
- IAM reviews

Resiliency Testing

- Zone-failure testing
- Regional failover testing
- DR exercises

Cutover Rehearsals

- Full production simulations
- Rollback rehearsal

16. Risk Register

Risk	Impact	Mitigation
Hidden Dependencies	Schedule delays	Discovery workshops and dependency mapping
Legacy .NET Applications	Migration delays	Early assessment and proof-of-concept
Data Migration Issues	Data loss	Replication monitoring and rehearsals
Security Approval Delays	Release blockage	Security embedded from Day 1
Resource Constraints	Reduced throughput	Dedicated migration teams
Scope Creep	Timeline risk	Strict change control
DR targets not achieved	Program failure	Incremental DR testing

Each risk must have Owner, Mitigation plan, Due date, Risk severity (Red / amber / Green) and Escalation path.

17. KPIs and Reporting

Delivery Metrics

- Apps migrated vs plan
- Milestone adherence
- Schedule variance

Quality Metrics

- Sev-1 / Sev-2 incidents
- Rollback count
- Defect leakage rate

Reliability Metrics

- Availability
- SLO attainment
- RTO achievement
- RPO achievement

Financial Metrics

- Cloud spend vs forecast
 - Legacy cost reduction
 - Budget variance
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18. Executive Dashboard

A one-page dashboard reviewed weekly by leadership.

Delivery

- Applications Migrated: 14 / 20
- Current Wave Status: Green
- Milestones On Track: 5 / 6

Quality

- Sev-1 Incidents: 0
- Sev-2 Incidents: 1
- Rollbacks: 0

Risk

- Open Risks: 4
- Critical Risks: 1

Reliability

- Availability: 99.96%
- DR Drill Status: Passed

Financial

- Budget Variance: +2%
 - Legacy Infrastructure Retired: 70%
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19. Program Success Metrics

Objective	Success Metric
Migration Completion	20/20 Applications Migrated
Business Continuity	Zero Sev1 Incidents
Data Integrity	Zero Data Loss
Reliability	99.95% Availability
Disaster Recovery	RPO <= 5 min, RTO <= 60 min
Operational Adoption	SRE Handoff Completed
Cost Optimization	Legacy Platform Retired

20. Program Governance and TPM Responsibilities

The TPM is accountable for:

- Program planning and execution
- Milestone governance
- Dependency management

- RAID ownership
- Cross-functional coordination
- Executive reporting
- Risk and issue management
- Operational readiness and SRE handoff

21. Hypercare Exit Criteria

Hypercare can close only when:

- No Sev1 incidents for 14 consecutive days
 - No Sev2 incidents for 7 consecutive days
 - SLOs meeting targets
 - Defects below threshold
 - SRE accepts ownership
 - Runbooks validated
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22. Appendix A - High-Level RACI

Activity	TPM	Platform	Security	App Team	DBA	SRE
Planning	A	C	C	C	C	C
Landing Zone	C	R	A	I	I	I
Dependency Mapping	A	C	I	R	C	I
Testing	C	C	I	R	C	C
Cutover	A	R	C	R	R	R
Hypercare	C	C	I	R	C	A
SRE Handoff	C	I	I	I	I	A

Legend:

- R = Responsible
 - A = Accountable
 - C = Consulted
 - I = Informed
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23. Non-Negotiables

1. No workloads migrate before the cloud foundation is approved.
2. Pilot wave must validate the entire migration process before scaling.
3. Every migration wave requires a tested rollback plan.
4. Disaster recovery is validated throughout the program, not only at the end.
5. No Sev-1 or Sev-2 production issues remain open before wave closure.
6. Executive stakeholders receive transparent status updates throughout the program.
7. The RAID log remains the single source of truth for risks, issues, dependencies, and decisions.